

Seat No:

Enrollment No:

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech, Winter 2025-26 Examination

Semester: V / VI

Subject Code: 303105307/203105322

Subject Name: Artificial Intelligence

Date: 24-11-2025

Time: 2.00 pm to 4.30 pm

Total Marks: 60

Instructions:

1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1** is compulsory, From Section B, **Q.1** is compulsory.
3. Figures to the right indicate full marks
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

SECTION - A

Q.1	Answer the following questions.	Marks	CO	BT
	A. (A) Define breadth-first search. (B) Define depth-first search. (C) Compare their use in AI.	06	CO1	BT1
	B. (A) Define frame representation. (B) List its components. (C) Explain how it is used in AI.	06	CO1	BT5
Q.2	A. State the basic concept of Predicate Logic and its role in knowledge representation.	04	CO1	BT1
	B. Define Bayesian Network and explain its components.	05	CO1, CO3	BT2
	OR			
	B. List the primary features of a Hidden Markov Model used in sequence prediction.	05	CO2	BT1
Q.3	A. Explain the concept of Backtracking search in AI with an example.	04	CO4	BT1
	B. Define and explain the importance of fuzzy set theory in knowledge-based systems.	05	CO3	BT2
	OR			
	B. State the concept of fuzzy set complement and explain its properties.	05	CO4	BT1

SECTION - B

Q.1	Answer the following questions.	Marks	CO	BT
	A. (A) Define AI and its major areas. (B) List AI techniques with examples. (C) Justify the importance of AI in modern society.	06	CO1	BT2
	B. (A) State the key properties of internal representation. (B) List the types of internal representations. (C) Justify their role in AI systems.	06	CO2	BT2

Q.2	A. List the different types of intelligent agents and explain the use of autonomous agents in AI.	04	CO1	BT1
	B. List the applications of AI in the medical field and explain the impact of AI in medical diagnostics.	05	CO1	BT1
	OR			
	B. Explain the importance of activation functions in neural networks and list common examples.	05	CO4	BT2
Q.3	A. Define first-order logic and give an example.	04	CO1	BT1
	B. Define word sense disambiguation and explain its significance in semantic analysis.	05	CO1	BT2
	OR			
	B. List the main components of Natural Language Processing.	05	CO4	BT1